

The Security of Machine Learning

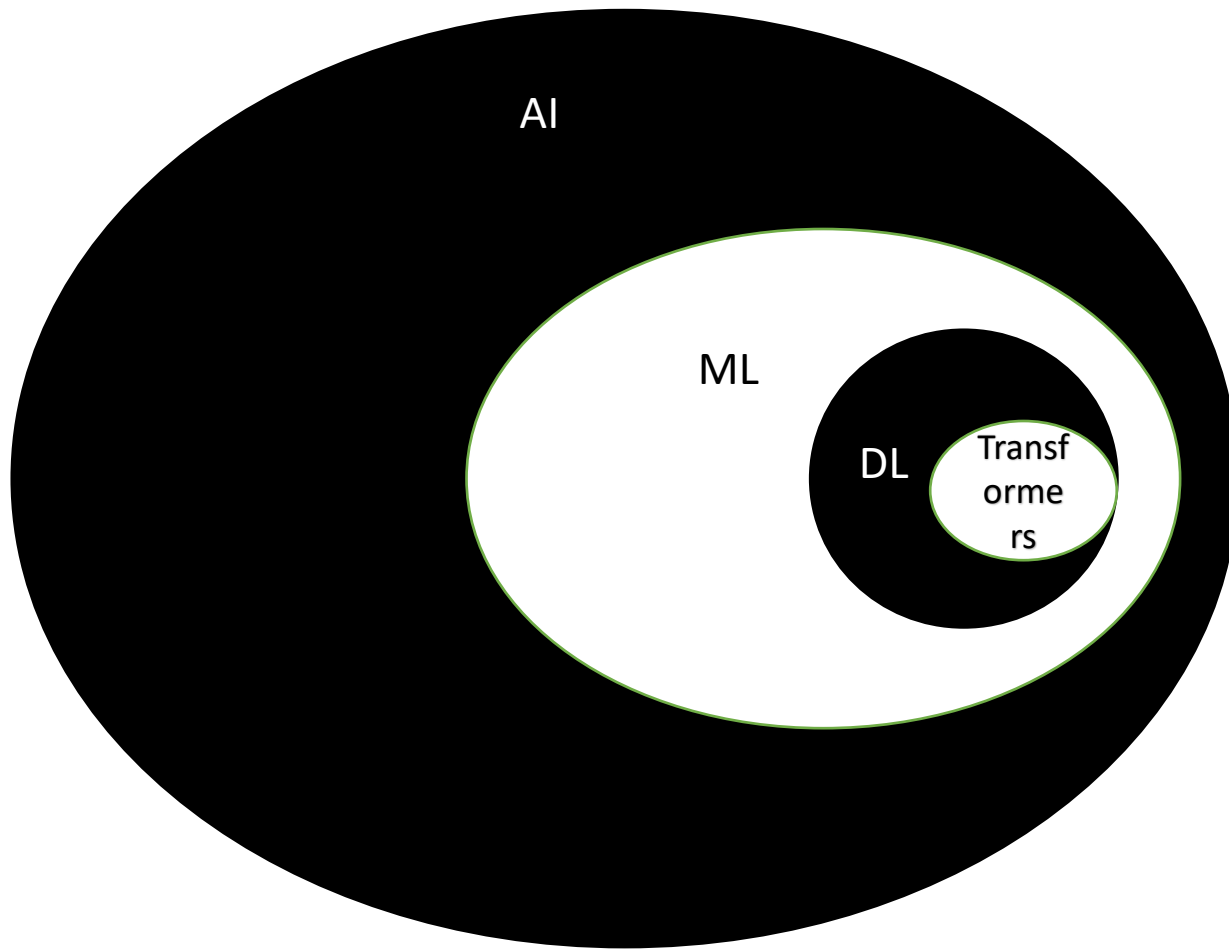
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FEMS, UoP, August'26

Various parts in collaboration with S. Liu, M. Santos, V. Gallego, R. Naveiro, PG Arce, F. Ruggeri, J. Poulos, S. Das

- ML in applied settings
- AML: An overview
- Bayesian AML
 - Bayesian approaches to AML
 - Bayesian approaches to ABML
- The actual problem!!
- Discussion

From the EU AI Act



distinguish it from simpler traditional software systems or programming approaches ...

*A key characteristic of **AI systems** is their capability to infer.*

*This capability to infer refers to the process of **obtaining the outputs, such as predictions, content, recommendations, or decisions,***

*which **can** influence **physical and virtual environments**, and to a capability of **AI systems** to derive models or algorithms, or both, from inputs or data.*

*The **techniques that enable inference while building an AI system** include **machine learning approaches** that learn from data how to achieve certain objectives, and **logic- and knowledge-based approaches** that infer from encoded knowledge or symbolic representation of the task to be solved.*

*The capacity of an AI system to infer transcends basic data processing by **enabling learning, reasoning or modelling...***

ML in applied settings

ML in Finance

- **Fraud detection** – Identifies suspicious credit card or banking transactions by recognizing unusual spending patterns.
- **Credit risk assessment** – Predicts the likelihood that a borrower will default on a loan using historical financial data.
- **Algorithmic trading** – Uses ML models to identify trading opportunities and execute buy/sell orders automatically.
- ...

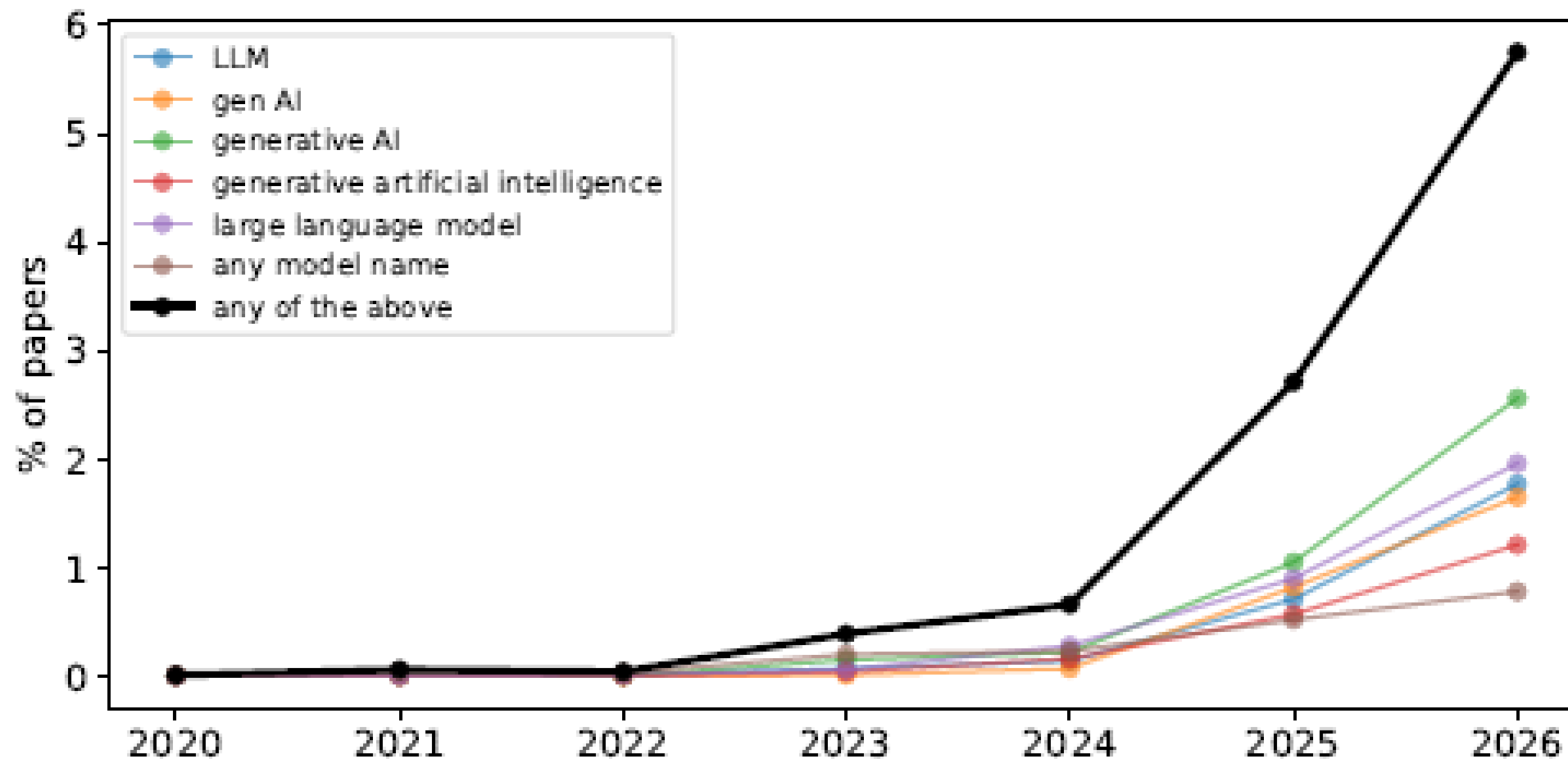
Economics

- **Economic forecasting** – Predicts GDP growth, inflation, or unemployment using macroeconomic data.
- **Consumer behavior analysis** – Identifies patterns in spending and saving habits to understand demand.
- **Policy impact evaluation** – Estimates the effects of government policies (e.g., tax changes or subsidies) on economic outcomes.
- ...
- .

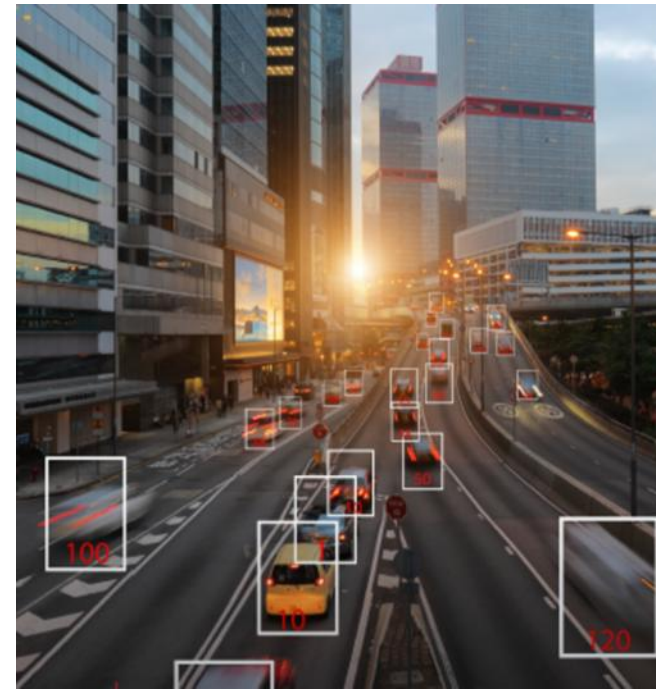
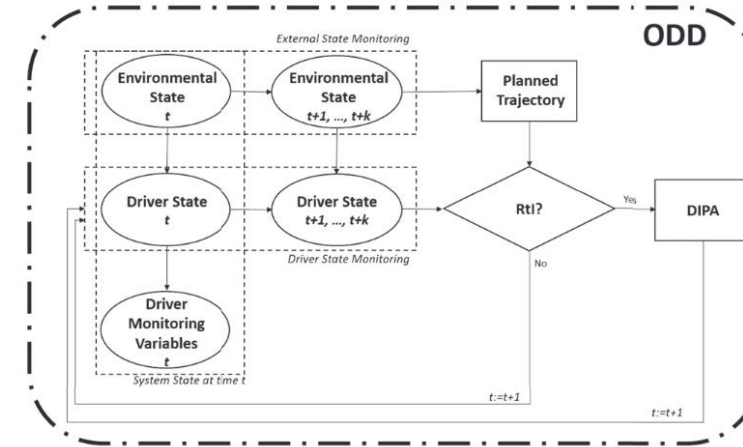
Business analysis

- **Sales forecasting** – Predicts future sales using historical data, seasonality, and market trends.
- **Demand forecasting** – Estimates future product demand to improve inventory and supply chain planning.
- **Product launching** Supports PL decisions taking into account competitors and market structure
- ...

LLMs in Business and Econ Research (Coqueret et al 2026)



ADS



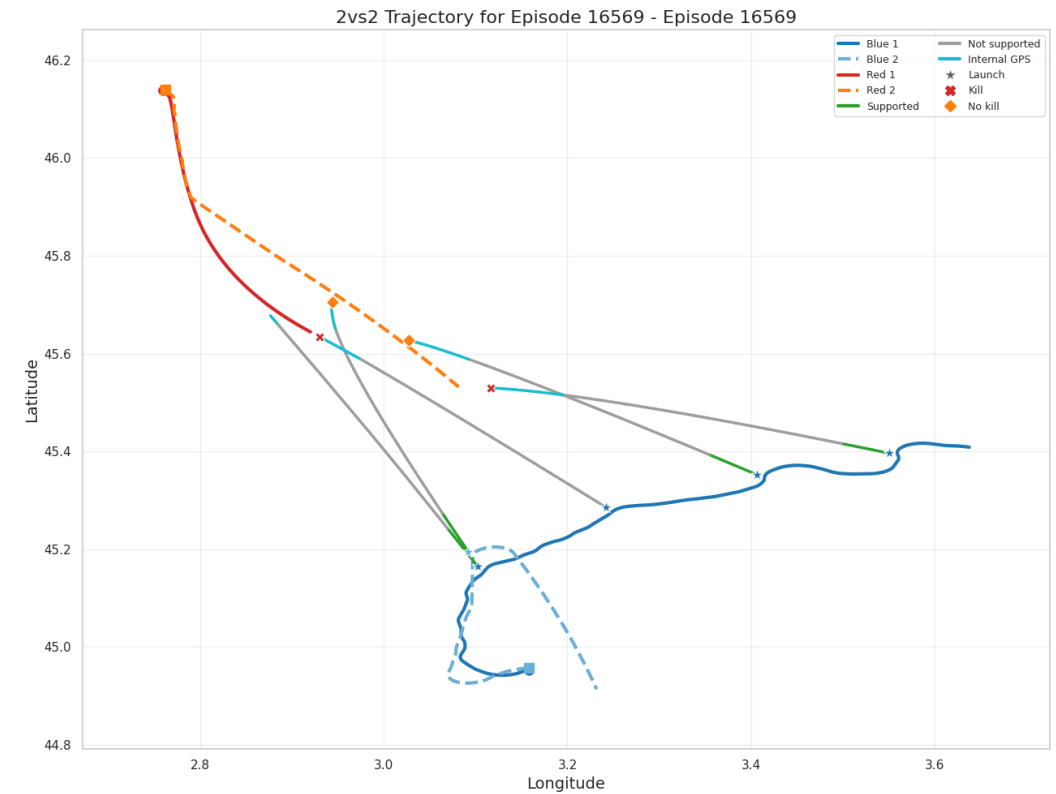
H2020 Trustonomy

DRI, Caballero, Naveiro (2022) Transp. Science

Air combat trainers



AirbusD
Hybrid: PID Control+RL

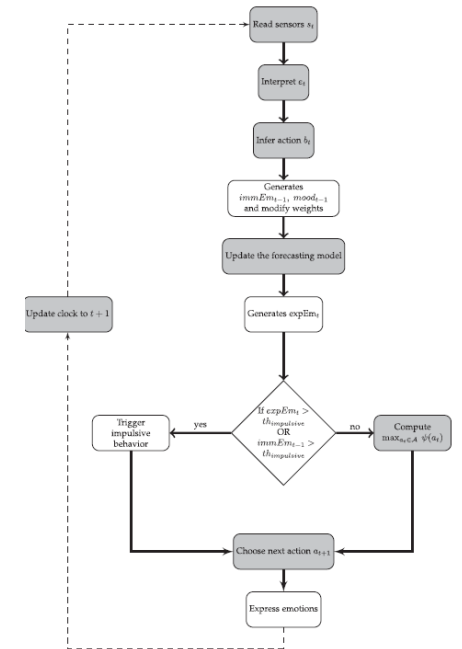


Social robots



EMOROBCARE. SEDIA

Esteban, Liu, DRI, Ortega (2020) Auton. Robots



Cybersec and AI as global risks. WEF GRM (2024)

2 years	10 years
1 st Misinformation and disinformation	1 st Extreme weather events
2 nd Extreme weather events	2 nd Critical change to Earth systems
3 rd Societal polarization	3 rd Biodiversity loss and ecosystem collapse
4 th Cyber insecurity	4 th Natural resource shortages
5 th Interstate armed conflict	5 th Misinformation and disinformation
6 th Lack of economic opportunity	6 th Adverse outcomes of AI technologies
7 th Inflation	7 th Involuntary migration
8 th Involuntary migration	8 th Cyber insecurity
9 th Economic downturn	9 th Societal polarization
10 th Pollution	10 th Pollution

StCrMg ERR

<https://scientificadvice.eu/advice/strategic-crisis-management-in-the-eu/>



International AI Safety Report



From the EU AI Act. Risk (617)

*In order to ensure a consistent and high level of protection of public interests as regards health, safety and fundamental rights, **common rules for high-risk AI systems should be established.***

In order to introduce a proportionate and effective set of binding rules for AI systems, **a clearly defined risk-based approach should be followed.** That approach should tailor the type and content of such rules to the intensity and scope of the risks that AI systems can generate. It is therefore necessary to prohibit certain **unacceptable** AI practices, to lay down requirements for high-risk AI systems and obligations for the relevant operators, and to lay down transparency obligations for certain AI systems.

GDPR for XAI. EUAIAct for SecAI?

Some ML examples. Red matters!!!

Uncertainty is almost ubiquitous in ML:

- Given a certain transaction, is it fraudulent or not? **If fraudulent should I stop it?**
- Given the monitoring trace of an Inet device, are we facing an attack? **Should I stop operations?**
- A person with these FB likes will buy this type of beer? **Should I send him my brand add?**
- A person with these tweets is conservative? **Should I send her Brexit propaganda?**
- Robots (or ADS): If robot performs this, How will the user react? And the environment? **Consequently, what should the robot do?**

ML meets sec

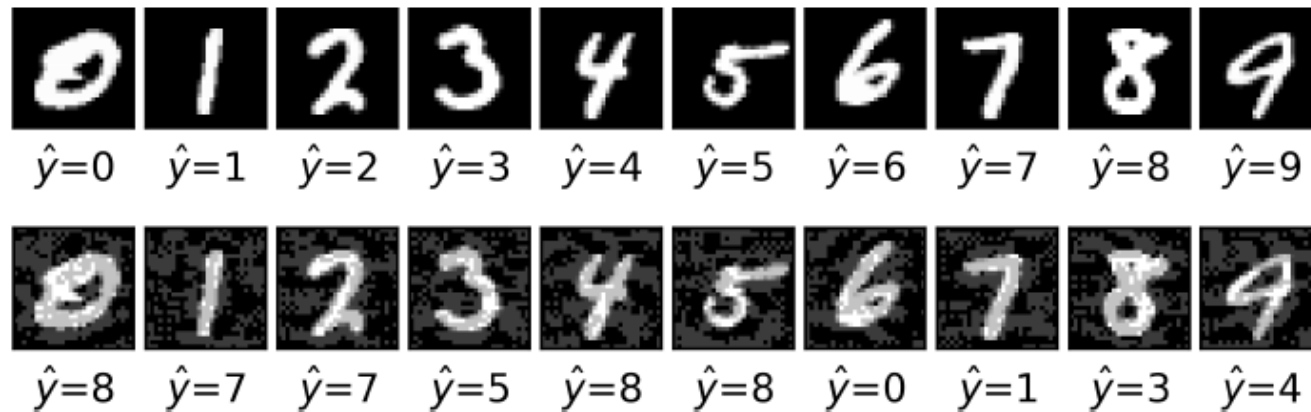
ML meets sec

Traditional assumption in ML: iid data

Training and operational data generated from same distribution

AML Attacks

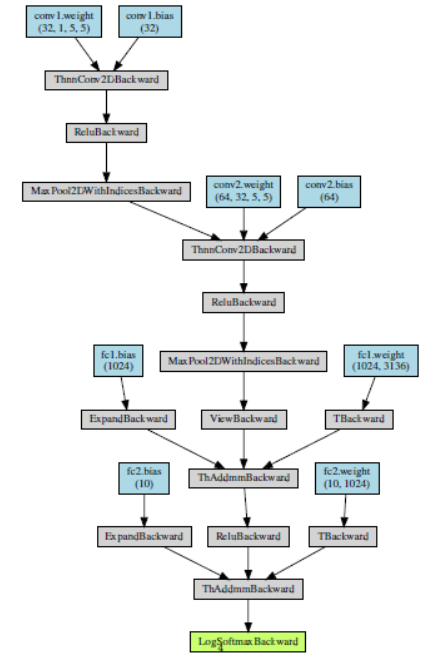
Attacks over data



Accuracy 99%

With attacked data 62%

FGSM Fast Gradient Sign Method $x' = x + \epsilon \text{sign}(\nabla_x L(x, y, \theta))$

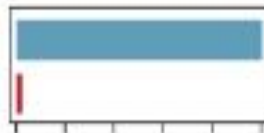


So what?....

Original image

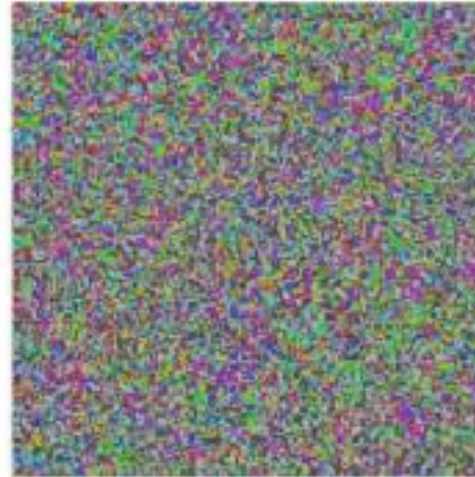


Dermatoscopic image of a benign melanocytic nevus, along with the diagnostic probability computed by a deep neural network.



Benign
Malignant

Adversarial noise

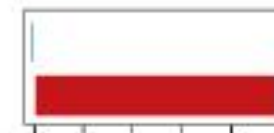


Perturbation computed by a common adversarial attack technique. See (7) for details.

Adversarial example

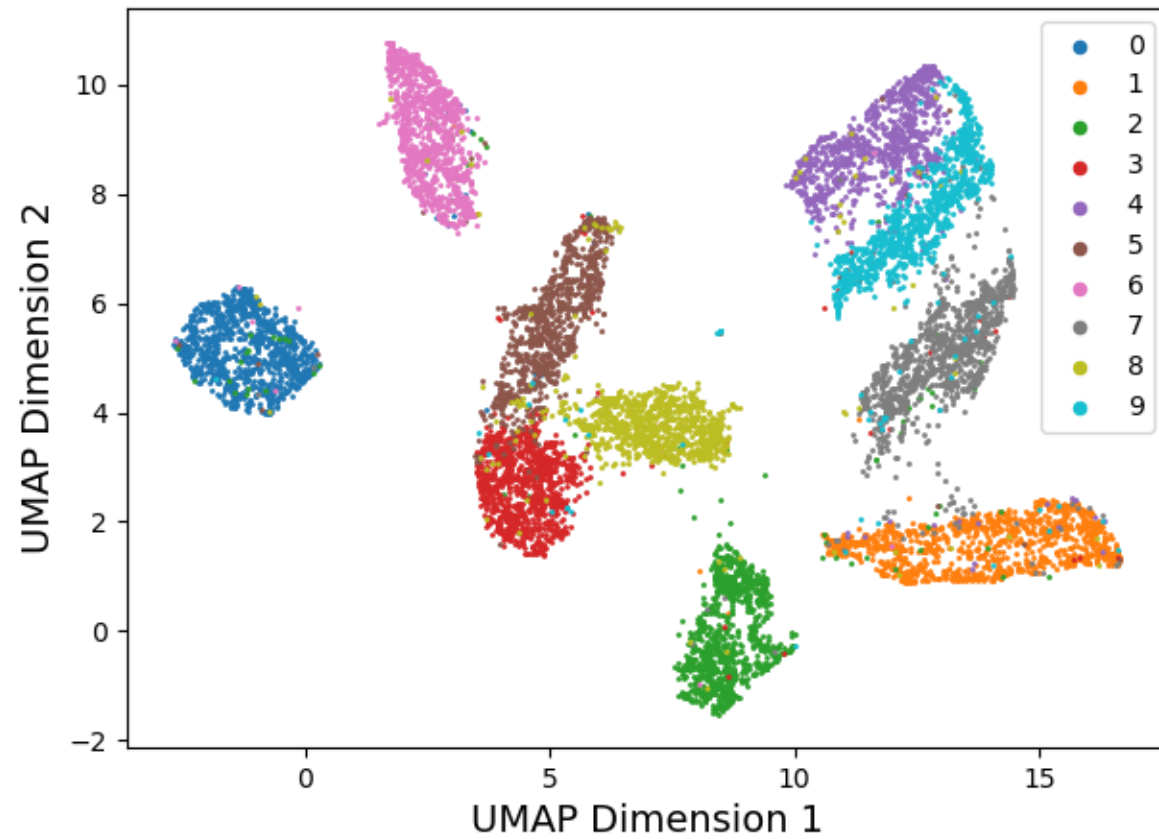


Combined image of nevus and attack perturbation and the diagnostic probabilities from the same deep neural network.

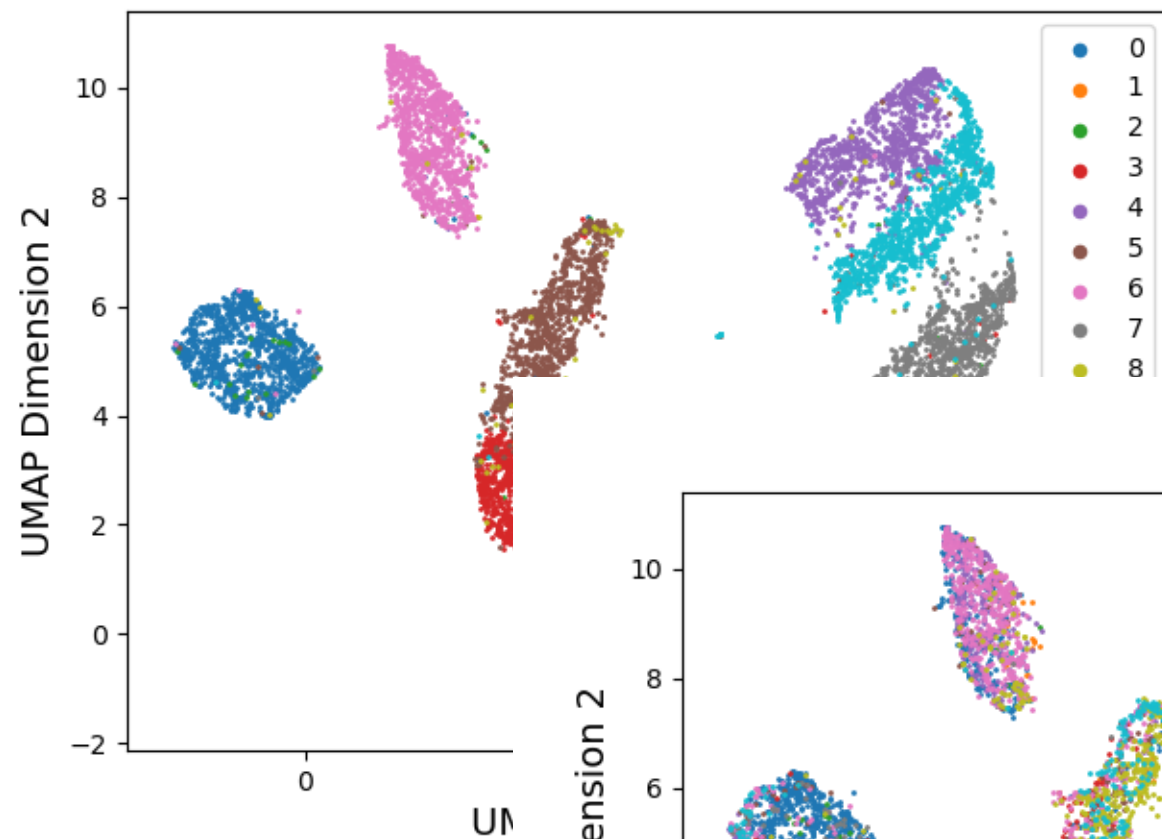


Benign
Malignant

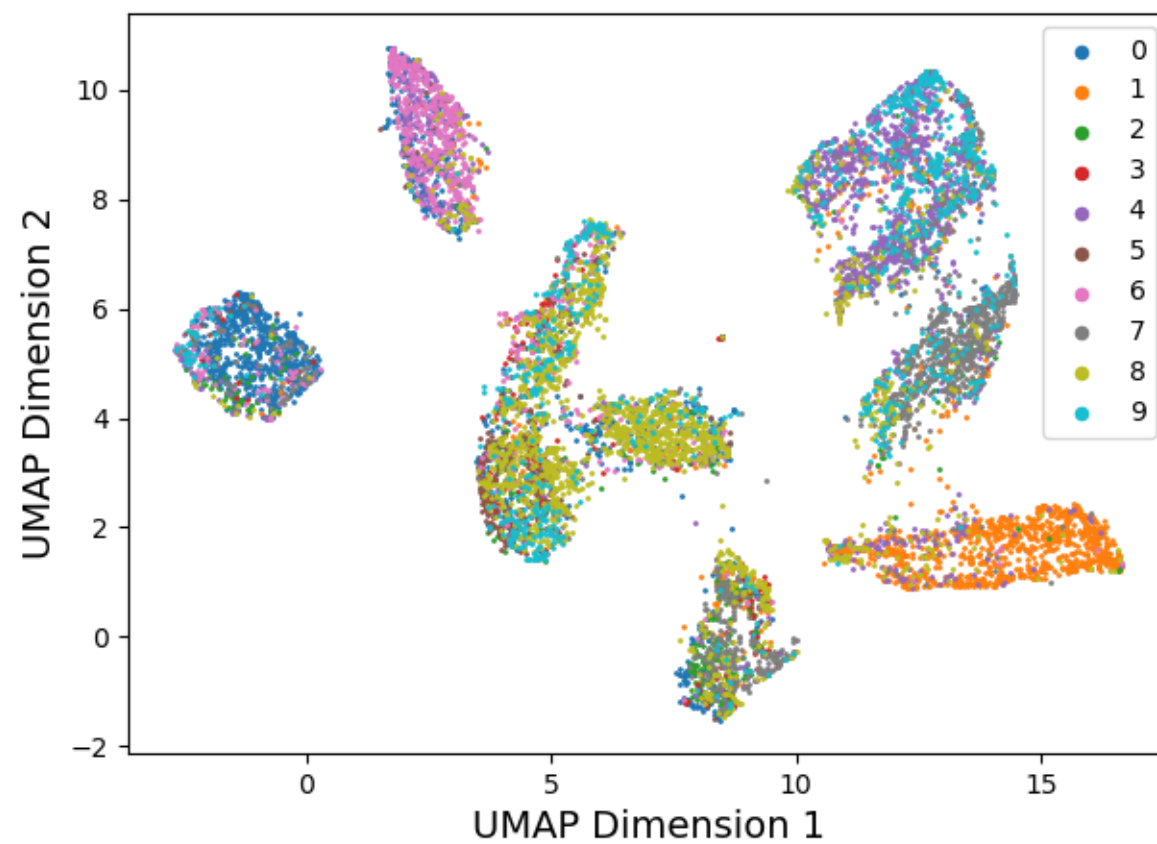
Source: Finlayson et al. (2019)



PG Arce, DRI (2025)



PGD



ML Attacks

Attacks over different types of data and different types of algos

Images Gallego et al (2023) JASA

Text Naveiro et al (2019) IJAR

Numbers PG Arce et al (2024) Wiley StatsRef

Sound https://nicholas.carlini.com/code/audio_adversarial_examples/

LLMs <https://arxiv.org/abs/2307.15043>

Classifier	Acc. Unt.	Acc. Taint.
Naive Bayes	0.891 ± 0.003	0.774 ± 0.026
Logistic Reg.	0.928 ± 0.004	0.681 ± 0.009
Neural Net	0.905 ± 0.003	0.764 ± 0.007
Random Forest	0.946 ± 0.002	0.663 ± 0.006

Defences

- **Reactive defenses**
 - Timely detection of attacks
 - Frequent ML retraining
 - Verify correctness of ML decisions through domain experts
- **Proactive defenses**
 - **Security-by-Design**

Explicitly account for adversarial manipulations.
Secure/robust learning, attack detection...
 - **Security-by-Obscurity**

Information hiding, randomization...

During training, During operations

Defences

Defences: Adversarial training (Madry et al, 2018)

Train model using attacked samples

Minimize empirical risk under worst case perturbation

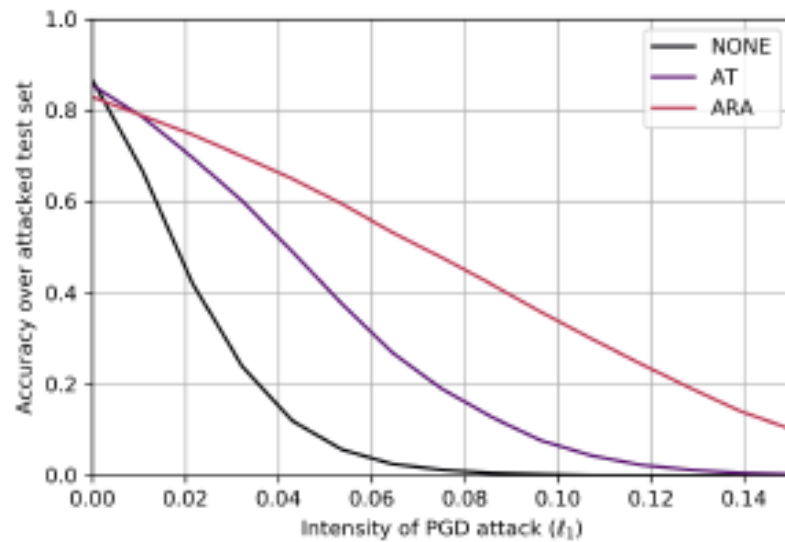
$$\min_{\theta} \mathbb{E}_{(x,y) \sim \mathcal{D}} \left[\max_{\|\delta_x\| \leq \epsilon} \hat{c}_D(h_{\theta}(a(x)), y) \right]$$

AT, ALP, TRADES,...

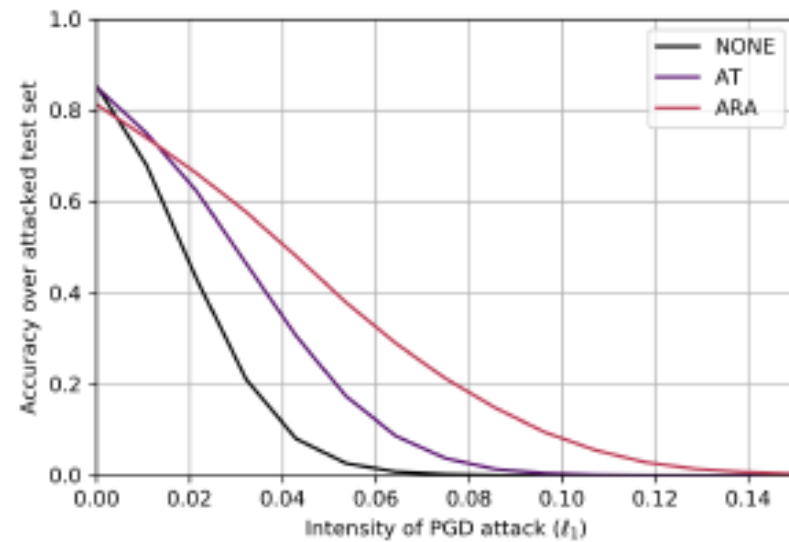
Purification: Randomised smoothing,...

Defences

Defences. Security curves



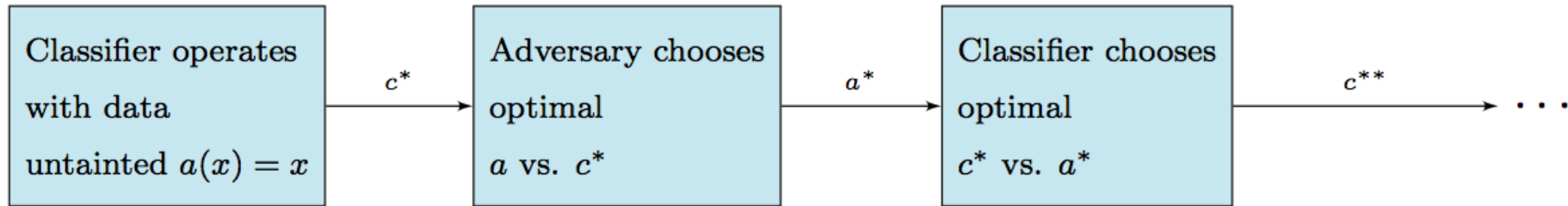
(a) Fashion-MNIST dataset.



(b) Kuzushiji-MNIST dataset.

AML Pipelines

Pipelines. Dalvi et al (2004)
(based on utility sensitive NB classifier)



Forward myopic approach under **strong common knowledge!**

AML SOTA

- Everything adversarial: adversarial SVM's,....
- Mostly viewed from a *game theoretic perspective*, over *MLE+reg approaches and point preds*
- Common knowledge too commonly assumed.
Uncertainty about attackers? Uncertainty about the uncertainty of attackers? Too deterministic
- A very difficult area
- Very relevant area from an applied point of view: security and cybersecurity

Methodological Flow From RA to ARA

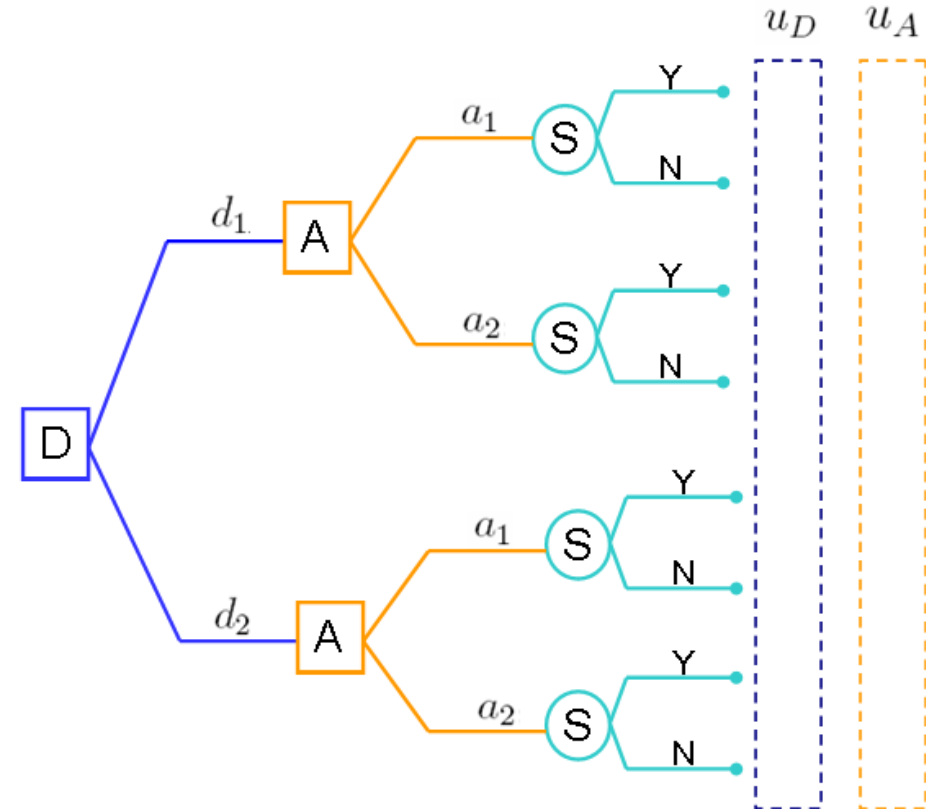
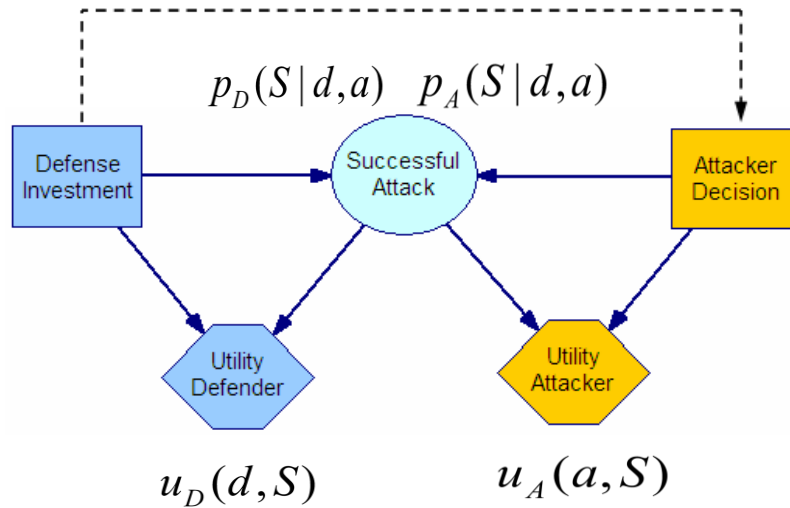
From RA to ARA...



Adversarial Risk Analysis

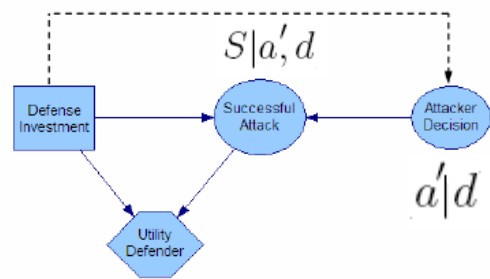
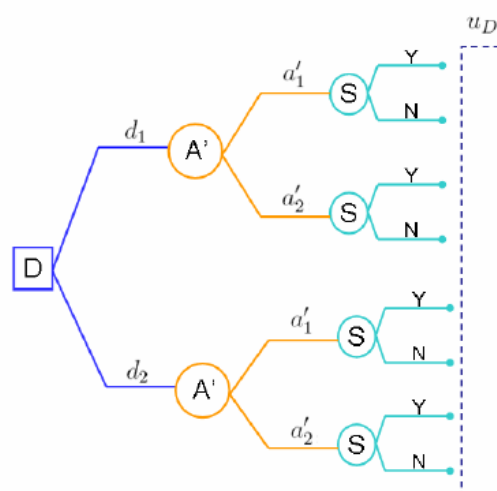
Sequential Defend-Attack DRI, Banks, Rios (2009) JASA,...

- Two intelligent players
 - Defender and Attacker. D knows A's judgements
- Sequential moves
 - Def, then Attacker



ARA

Defender problem



Defender's solution

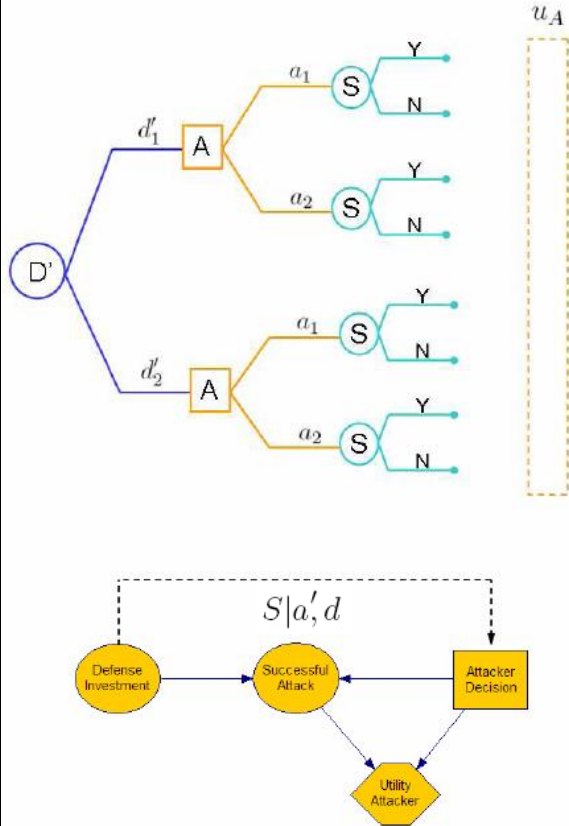
$$\psi_D(d, a') = u_D(d, S = Y) p_D(S = Y | X_D = d, X'_A = a') + u_D(d, S = N) p_D(S = N | X_D = d, X'_A = a')$$

$$\psi_D(d) = \psi_D(d, a'_1) p_D(a'_1 | d) + \psi_D(d, a'_2) p_D(a'_2 | d)$$

$$d^* = \arg \max_{d \in X_D} \psi_D(d)$$

Modeling input: $p_D(S|a', d)$ $p_D(a'|d)$??

ARA

Defender's view of Attacker problem	Elicitation of $p_D(a' d)$
 The diagram shows an extensive form game tree for the defender's view of the attacker's problem. The defender (D') chooses between two defense strategies, d'_1 and d'_2. For each strategy, the attacker (A) chooses between two actions, a_1 and a_2. For each action, the attacker's strategy (S) is either successful (Y) or unsuccessful (N). The utility for the attacker is u_A. Below the tree is a directed graphical model with four nodes: Defense Investment (yellow circle), Successful Attack (yellow circle), Attacker Decision (yellow rectangle), and Utility Attacker (yellow hexagon). Arrows point from Defense Investment to Successful Attack, from Successful Attack to Attacker Decision, and from Attacker Decision to Utility Attacker. A dashed arrow points from Attacker Decision back to Successful Attack, labeled $S a', d$.	A is a EU maximizer D's beliefs about $(\hat{u}_A, \hat{p}_A) \sim F$ $\hat{\psi}_A(d', a) = \hat{u}_A(a, S = Y) \hat{p}_A(S = Y X'_D = d', X_A = a) + \hat{u}_A(a, S = N) \hat{p}_A(S = N X'_D = d', X_A = a)$ $\hat{\psi}_A \sim \hat{\Psi}_A$ $p_D(a' d) = Pr\left[a' = \arg \max_{x \in X'_A} \hat{\Psi}_A(d, x)\right]$ <u>MC simulation</u> $\hat{p}_D(a d) \approx n^{-1} \sum_i \#\{a = \operatorname{argmax}_{x \in \mathcal{A}} \hat{\psi}_A^i(x, d)\}$ where $\hat{\psi}_A^i \sim \hat{\Psi}_A, i = 1, \dots, n$

Bayesian AML

Bayesian AML

- Replace standard game theoretic concepts with ARA (Bayesian DA approach to games) concepts for AML
- Securise Bayesian predictive models

A ‘foundational’ debate

- Bayesian predictive models are inherently adversarially robust (De Palma et al. 2021)
- Bayesian predictive models are not adversarially robust (Feng et al, 2024; Arce et al, 2025)

Bayesian approaches to AML

ACRA

Algorithm 1 ARA procedure for Adversarial Classification during operations

Input: N , training data $\mathcal{D}$.

Output: A classification decision $y_C^*(x')$.

Training

Based on $\mathcal{D}$ estimate a model for $p(y|x)$.

End Training

Operation

Read instance x'

Draw sample $\{x_n\}_{n=1}^N$ from $p(x|x')$.

Find $y_C^*(x') = \arg \max_{y_C} \frac{1}{N} \sum_{y=1}^k (u(y_C, y) \times [\sum_{n=1}^N p(y|x_n)])$

End Operation

Return $y_C^*(x')$

ACRA through ABC

Algorithm 2 MCMC-ABC sampler for $p(X|X' = x')$

Input: Instance x' , distribution $p(X'|X)$, prior $p(X)$, transition density $q(x \rightarrow \tilde{x})$ tolerance ϵ , set of statistics s and distance ϕ .

Output: Samples approximately distributed according to $p(X|X' = x')$.

Initialize $x_0, i = 0$.

Repeat until convergence:

Propose $\tilde{x}$ according to transition distribution $q(x_i \rightarrow \tilde{x})$.

Sample $\tilde{x}' \sim p(X'|X = \tilde{x})$.

Compute

$$\alpha = \min \left\{ 1, \frac{p(\tilde{x})q(\tilde{x} \rightarrow x_i)}{p(x_i)q(x_i \rightarrow \tilde{x})} \mathbb{I}[\phi(s(\tilde{x}'), s(x')) < \epsilon] \right\}$$

With probability α set $x_{i+1} = \tilde{x}$, otherwise $x_{i+1} = x_i$.

Set $i = i + 1$.

End Repeat

TABLE 3

Accuracy comparison (with precision) of four classifiers with and without protection on clean and attacked data. Attacker B

Classifier	Clean data (raw)	Attacked data (raw) Attacker B	Attacked data (AB-ACRA) Attacker B
Logistic Regression	0.728 ± 0.005	0.418 ± 0.010	0.840 ± 0.010
Naive Bayes	0.722 ± 0.004	0.405 ± 0.009	0.880 ± 0.027
Neural Network	0.691 ± 0.019	0.417 ± 0.015	0.700 ± 0.064
Random Forest	0.720 ± 0.005	0.397 ± 0.013	0.826 ± 0.012

ACRA training

Algorithm 5 ARA procedure for Adversarial Training of Differentiable Classifiers

Input: training data $\mathcal{D}$, prior $p(\beta)$.

Output: A classification decision $y_C^*(x')$.

Training

Use Algorithm 4 to obtain an approximation of $p(\beta|\mathcal{D})$ (robustified posterior).

End Training

Operation

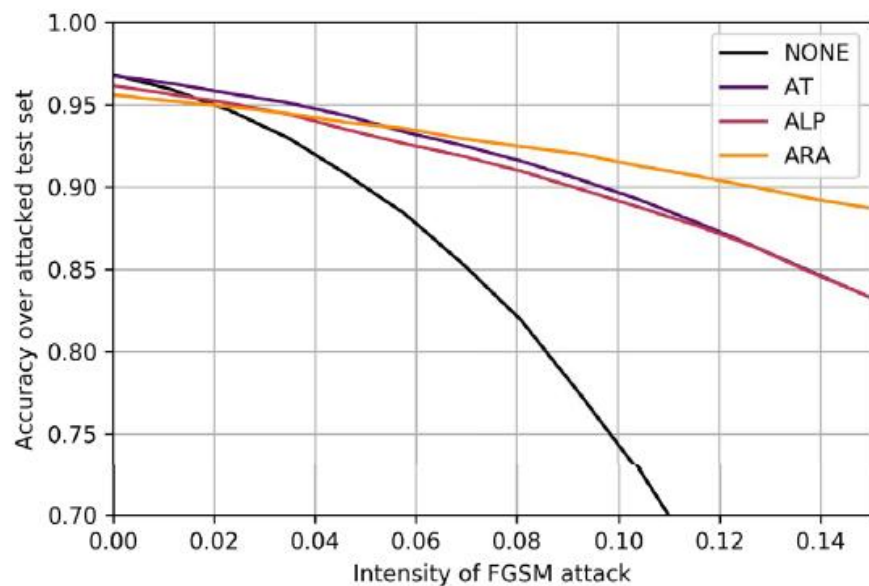
Read instance x'

Find $y_C^*(x') = \arg \max_{y_C} \sum_{y=1}^k u(y_C, y) \int p(y|x', \beta) p(\beta|\mathcal{D}) d\beta$

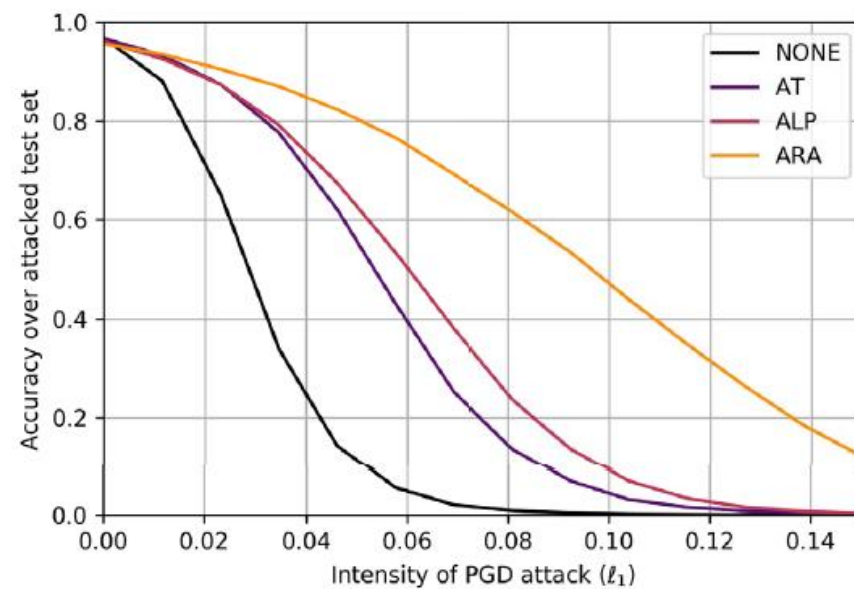
End Operation

return $y_C^*(x')$

ACRA-training



(a) FGSM attack.



(b) PGD attack under ℓ_1 norm.

Bayesian approaches to BML

Bayesian inherently adversarially robust?

Predictor: Bayesian model with posterior predictive distribution (PPD)

$$\pi(y|x, D) = \int \pi(y|f_{\beta}(x), \phi) \pi(\gamma|D) d\gamma, \quad \gamma \equiv (\beta, \phi)$$

Attacker: Seeks to manipulate inputs $x \rightarrow x'$ to achieve some objective.

Two Attack Types

1. **Point Attacks:** Target specific predictions (mean, quantiles, utilities,...)

$$\min_{x' \in \mathcal{X}} \|\mathbb{E}[g(x', y)] - G^*\|_2$$

2. **Distribution Attacks:** Reshape the entire PPD

$$\min_{x' \in \mathcal{X}} \text{KL}(\pi_A(y) \parallel \pi(y|x', D))$$

Targeting the full PPD

Objective: Steer PPD towards adversarial distribution $\pi_A(y)$

$$\min_{x' \in \mathcal{X}} \text{KL}(\pi_A(y) \parallel \pi(y|x', D))$$

Proposition (2)

Under some regularity conditions, the gradient can be expressed as:

$$\nabla_{x'} \text{KL} = -E_y \left[\frac{E_{\gamma|D}[\nabla_{x'} \pi(y|x', \gamma)]}{E_{\gamma|D}[\pi(y|x', \gamma)]} \right]$$

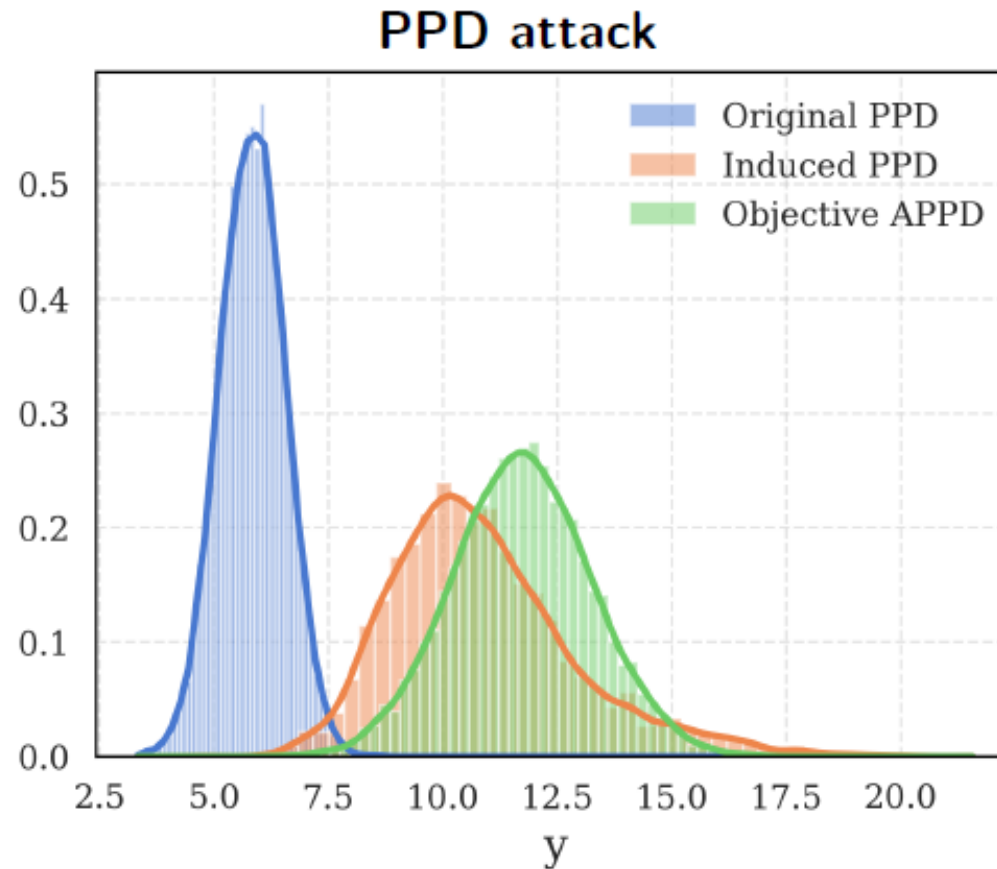
Challenge: Gradient involves ratio of expectations.

Solution: Multi-level Monte Carlo for unbiased estimation.

Targeting the full PPD

With $\|x' - x\|_2 \leq 0.5$

- Dataset: Wine quality. 11 features



Unifying Bayesian adversarially robust inference

Predictor: Bayesian model with posterior predictive distribution (PPD)

$$p(y|x, D) = \int p(y|x, \gamma) p(\gamma|D) d\gamma$$

Attacker: Seeks to manipulate inputs $x \rightarrow x'$ to achieve some objective.

Two Defense Types

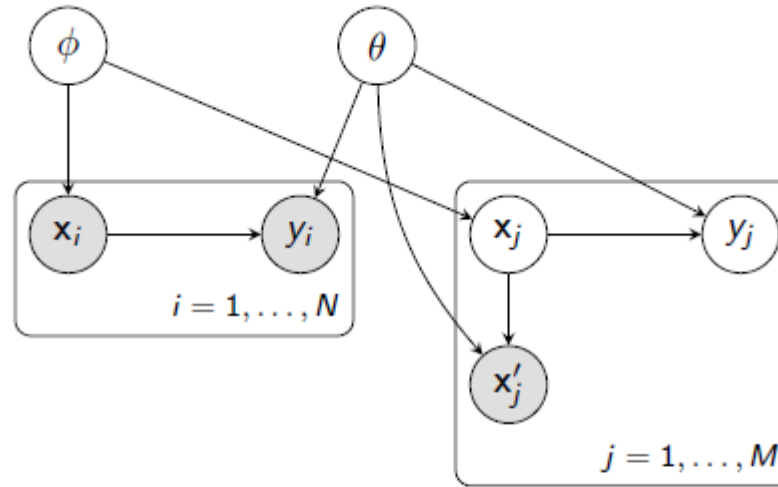
1. **Proactive:** Modify training to get a robust-by-design model

$$p_R(\gamma|D)$$

2. **Reactive:** Modify inference to undo the attack

$$p_R(y|x, D)$$

Reactive defense



- Assume **label invariance**: attack corrupts $\mathbf{x}_j \rightarrow \mathbf{x}'_j$, but $y_j \mid (\mathbf{x}_j, \theta)$ is unchanged.
- Defense inverts the adversarial channel to infer latent clean $\mathbf{x}_j$.

The actual problem!!

The actual problem

- Autonomous agents increasingly deployed in critical tasks
ADS, Defense, Social robots, Medicine, Investment,...
- Evolving in a dynamic context, interacting with other agents and persons
- Sense, Infer, Predict, Decide
(Negotiate, Compete, Protect, Defend)
- Goal: make their operations more secure

Formulation. Minum basic setup

- An autonomous system makes decisions a in an environment over time.
- Another system makes decisions b in the same environment.
- The environment adopts states θ
- The reference system has sensors capturing data x and y informing about b and θ
- Evaluation of a situation (a, b, θ) by the system is based on a utility function $u(a, b, \theta)$

Formulation. Basic setup

- Upon receiving x, y and planning a formulates a forecast

$$p(b, \theta | x, y, a) = p(\theta | x, y, a, b) p(b | x, y, a)$$

- Computes

$$\psi(a) = \int \int u(a, b, \theta) p(\theta | x, y, a, b) p(b | x, y, a) d\theta db$$

- Implements $a^* = \arg \max \psi(a)$ and expects to attain
 $\psi(a^*) = \psi^*$

NB: $u(a, b, \theta)$, $p(\theta | x, y, a, b)$ are standard in decision analytic terms but $p(b | x, y, a)$ is strategic.

Formulation. Adversarial issues

- An adversary messes with the problem and actually the system captures (x', y') .
- If system is unaware, will compute

$$\hat{\psi}(a) = \int \int u(a, b, \theta) p(\theta | x', y', a, b) p(b | x', y', a) d\theta db$$

- Implement $\hat{a} = \arg \max \hat{\psi}(a)$ and expect to attain $\hat{\psi}(\hat{a})$
- These could be quite different from a^* and ψ^* .

Formulation. Adversarial issues

- How to define attacks $x \rightarrow x'$ and $y \rightarrow y'$ so as to degrade system performance.
- How to protect system performance from such or other attacks at training and/or at operations?
- How to extend the above when there are several cooperating systems against several attackers

Formulation. Comments

- Standard AML focuses on alterations associated with (point) estimates of the probability part.
- Here focus is on whole distributions and, even more, the use of such distributions to find optimal alternative and expected utility.
- Normally only one type of attack is considered but we consider more than one attack. Attack surface is bigger (but somehow this robustifies the system)

Wrapping up

Some open issues

- Adversarial issues in RL
 - Adversarial issues in UL
 - Decision theoretic issues in AML
 - Scaling computations
-
- Banking: ARA and causal analysis (Bsantander)
 - Software environment
 - Security of social robots

A guided tour through AML

Dalvi et al (2004)

.....

Books

Vorobeychik, Kantarcioglu (2019)

Joseph, Nelson, Rubinstein, Tygar (2019)

Naveiro, DRI (2026)

Review Papers

Dasgupta, Collins (2019)

Biggio, Rolli (2018)

Zhou, Kantarcioglu, Xi (2021)

DRI, Gallego, Naveiro, Poulos (2023)

G.Arce, DRI (2024)

Policy

NIST (2021), Comiter (2019), EY (2022), EU AI Act (2023)

Some refs

- Insua et al (2023) JASA
- Gallego et al (2024) Stat Science
- Gallego et al (2027) EJOR
- Arce et al (2025) UAI
- Arce et al (2026) arxiv
- Liu et al (2026) arxiv
- Naveiro and DRI (2026) BAML, T&F

Thanks!!!

EU H2020: Starlight	(LEAs)
EU H2020: Trustonomy	(ADSs)
US Air Force, Airbus D	(Defence)
Caixabank, BBVA, B Santander	(Banking)
AXA RF	(Insurance)
SEDIA, Sp. Gvt	(Maths, CS)

Thanks!!!

Many open problems... collabs welcome

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